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Fang

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(54) **TELESCOPING GATE BRACE**

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Y10T 403/32491; Y10T 403/7084
USPC 403/109.4, 379.3; 256/30, 31, 35, 63, 64,
256/73; 248/354.1, 354.3, 354.6
See application file for complete search history.

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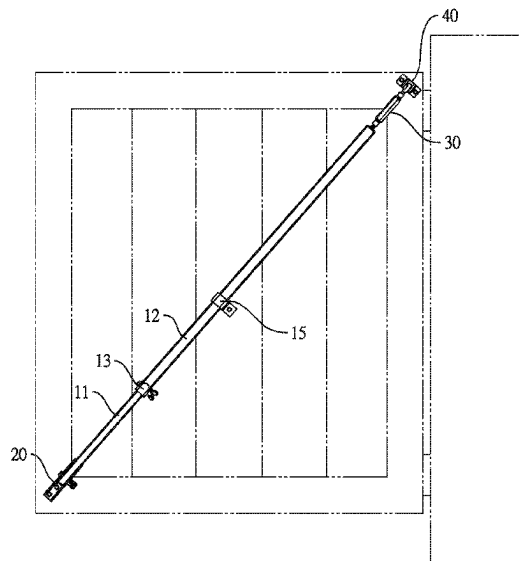
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(57)

ABSTRACT

A telescoping gate brace is configured to be mounted on a
wooden gate. The telescoping gate brace has a first rod, a
second rod, a securing component, a first fixing assembly, a
connector, and a second fixing assembly. The first rod and
the second rod are movably sleeved with each other. The
securing component is mounted on the first rod and the
second rod, thereby securing a total length of the first rod
and the second rod. The first fixing assembly is mounted on
the first rod and the wooden gate. The connector is mounted
on the second end. The second fixing assembly is mounted
on the connector and the wooden gate. A distance between
the second end and the second fixing assembly is adjustable
via the connector.

19 Claims, 7 Drawing Sheets



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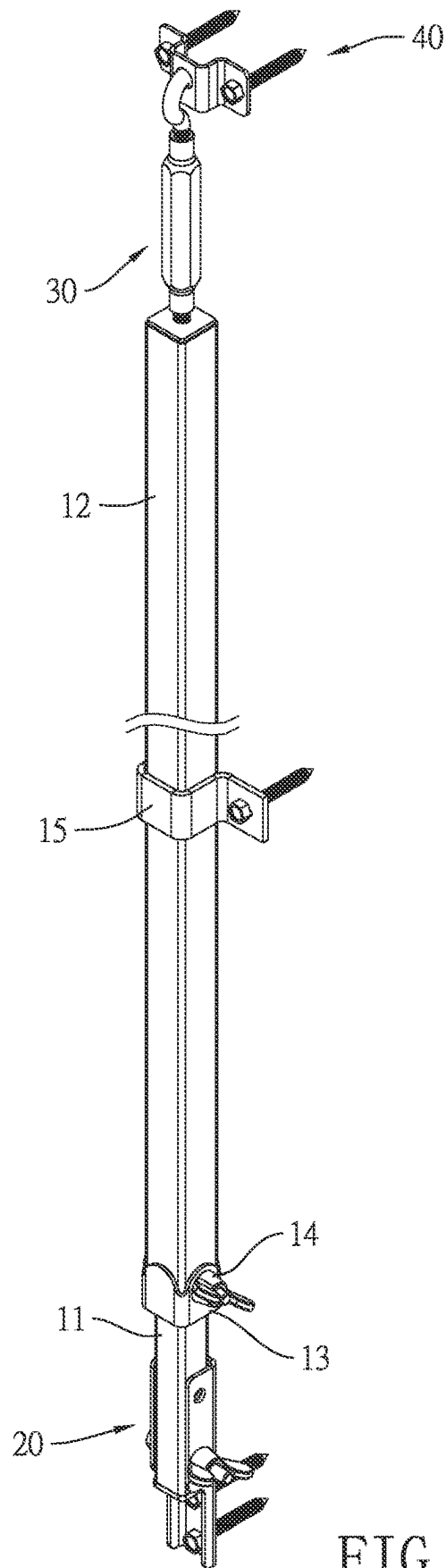


FIG. 1

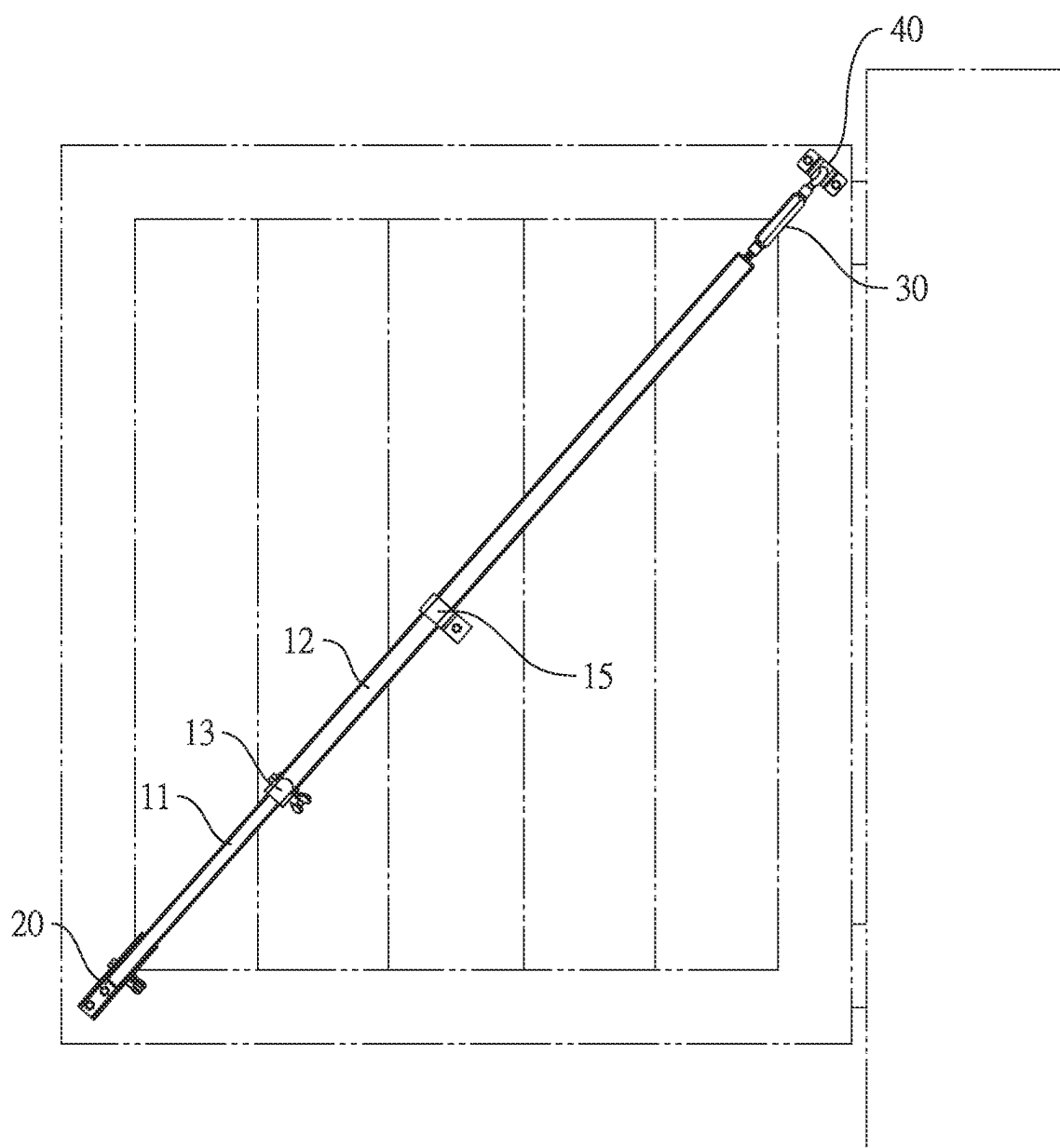


FIG. 2

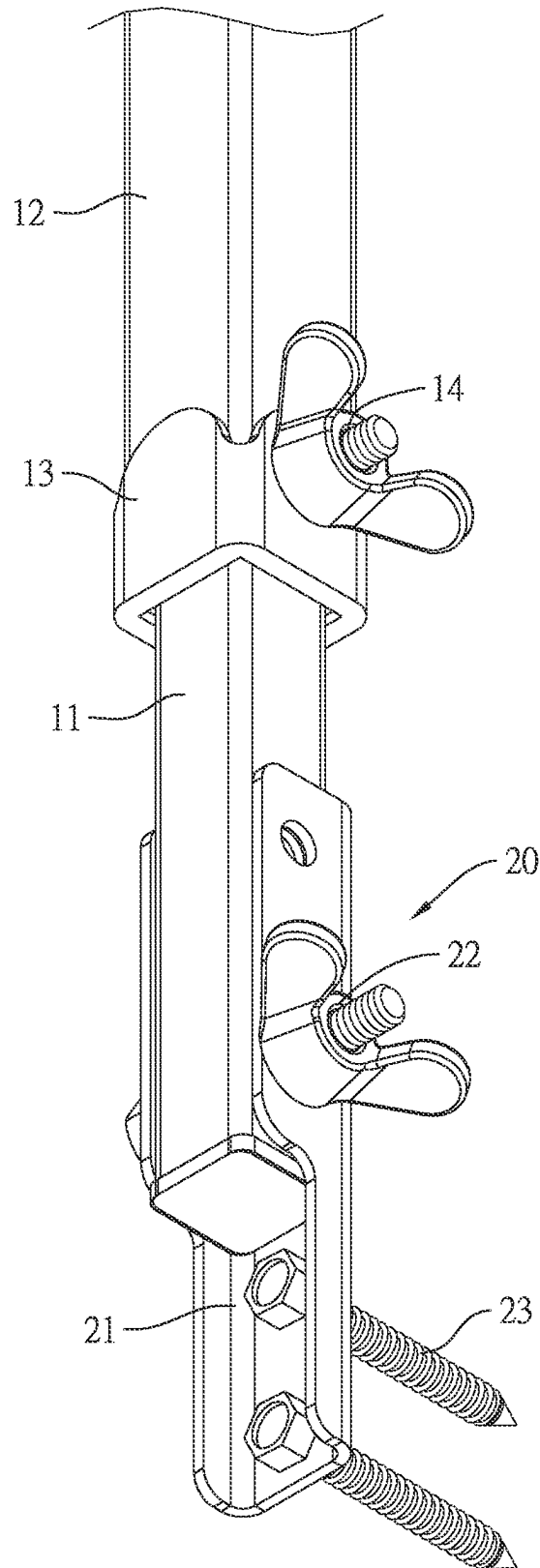


FIG. 3

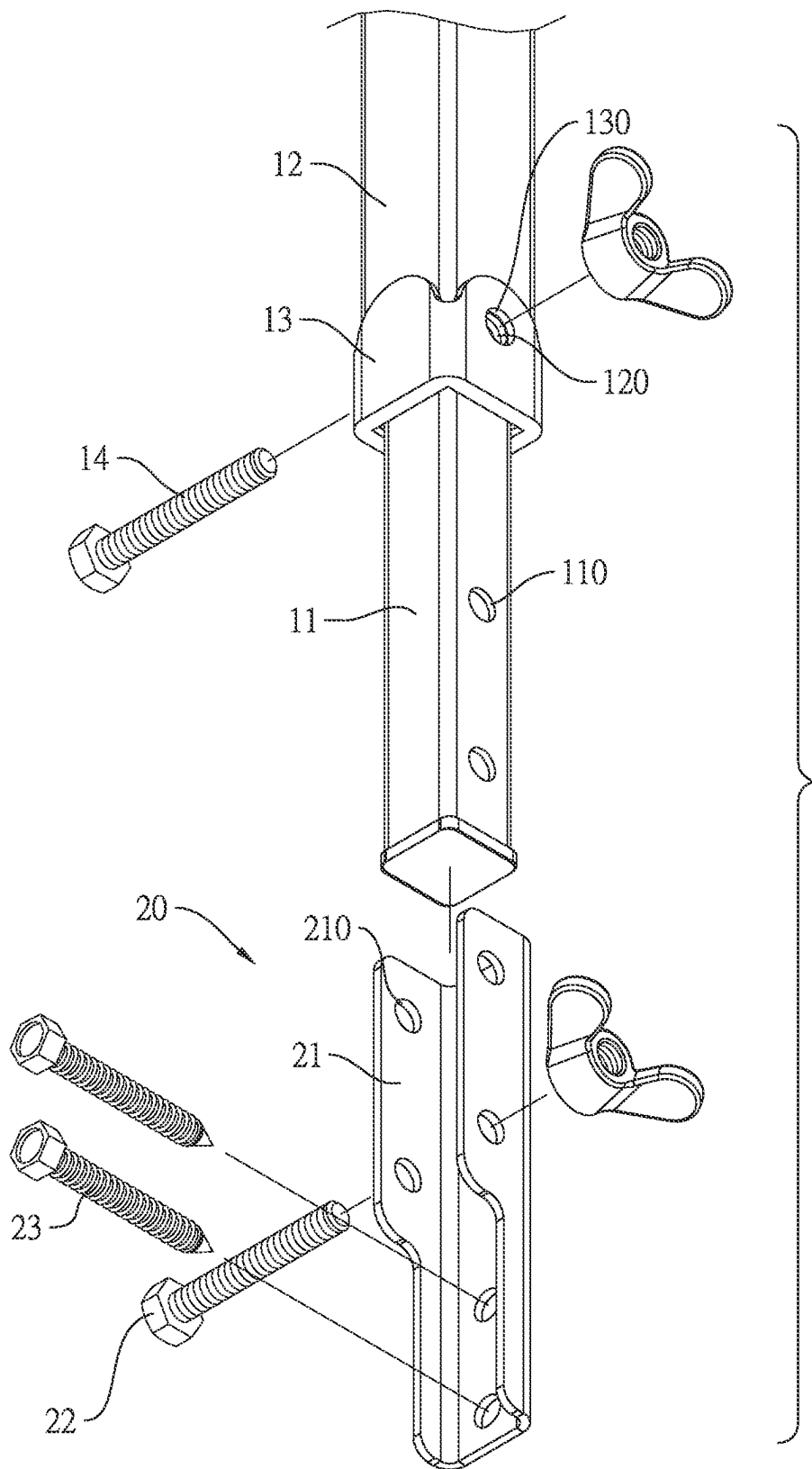


FIG. 4

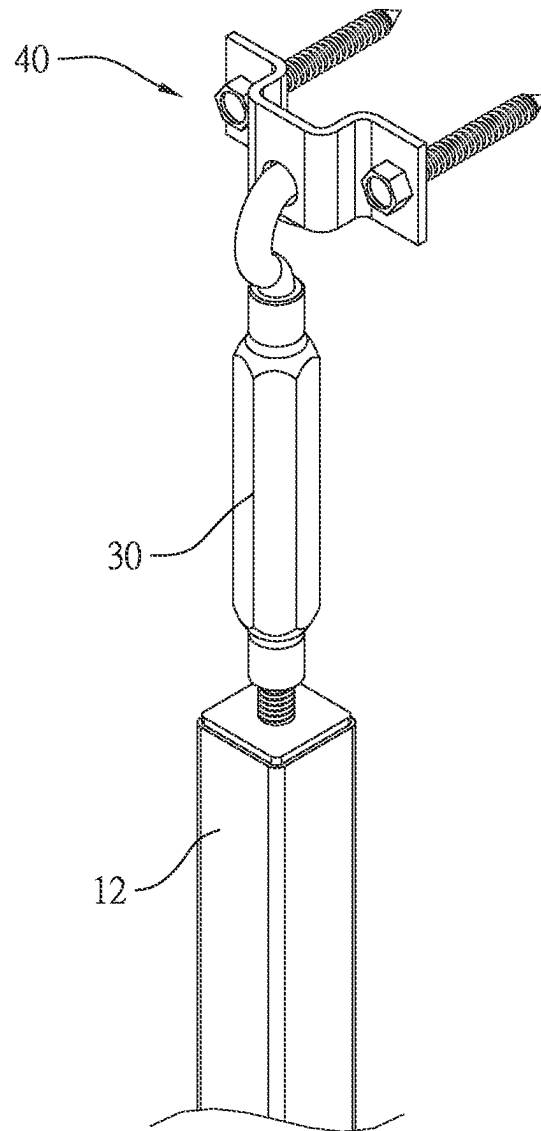


FIG. 5

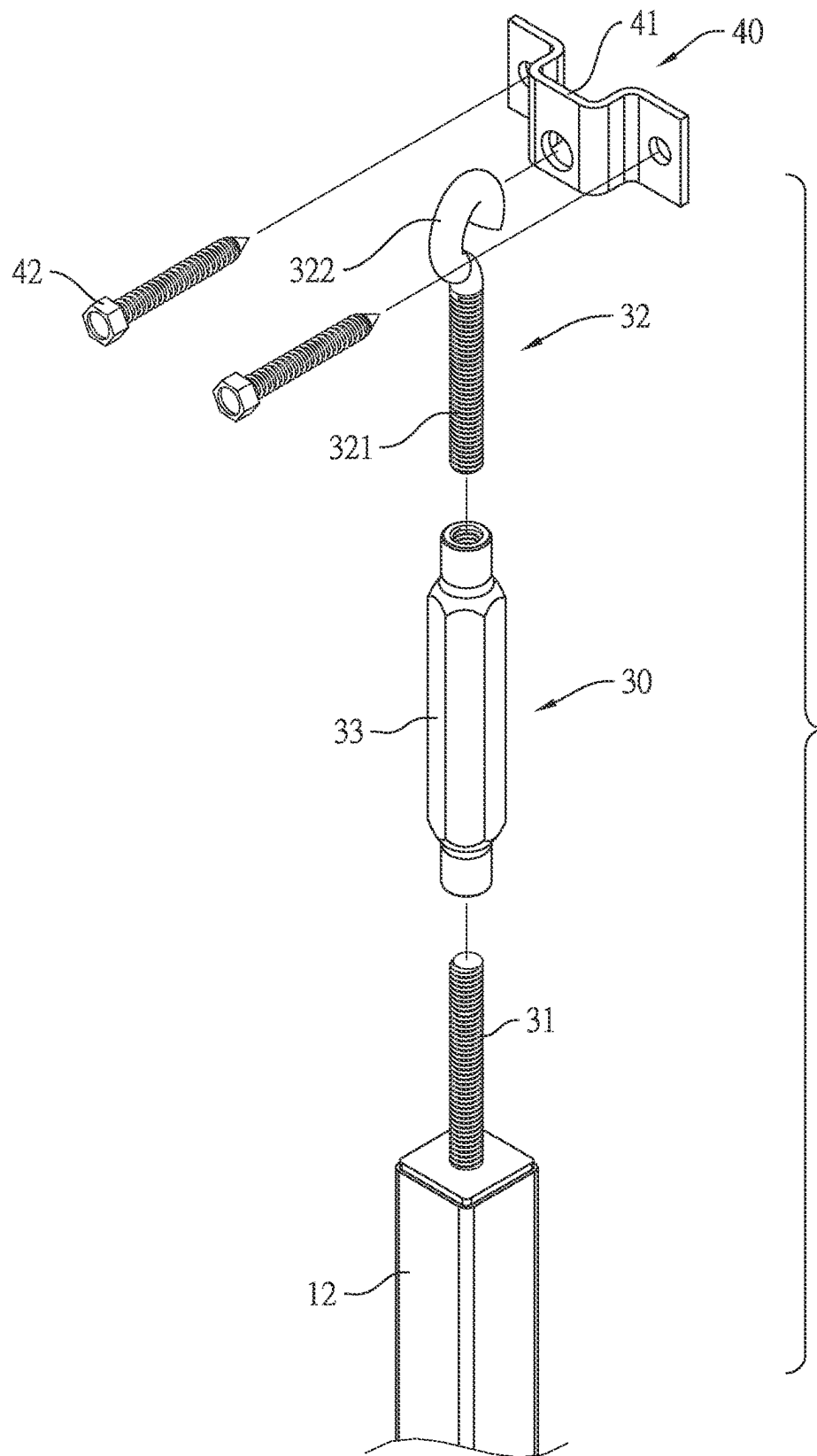


FIG. 6

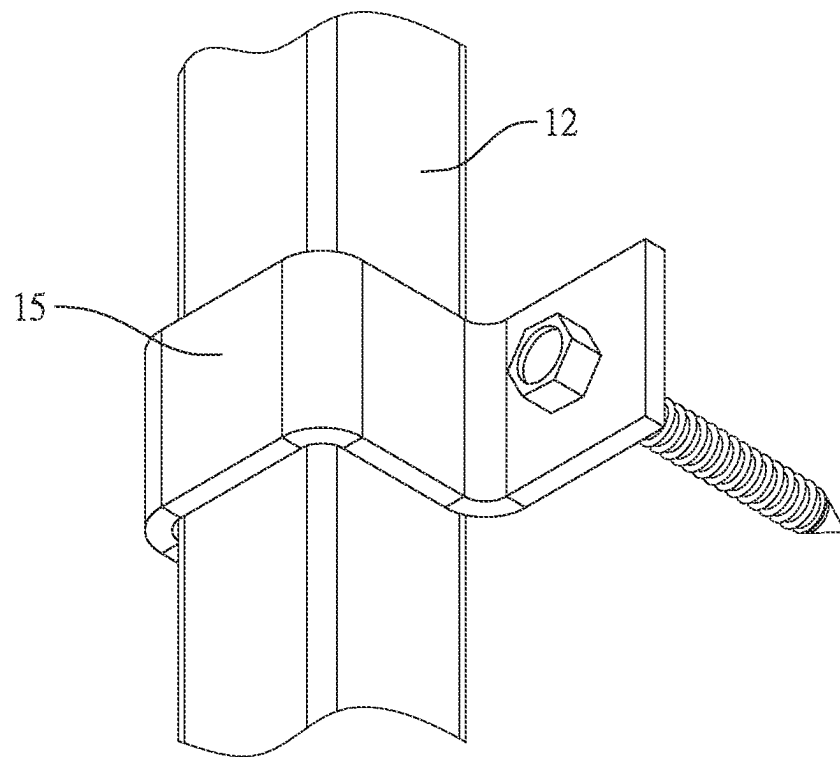


FIG. 7

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TELESCOPING GATE BRACE**BACKGROUND OF THE INVENTION****1. Field of the Invention**

The present invention relates to a telescoping gate brace, especially to a support mounted on a wooden gate to prevent the wooden gate from sagging.

2. Description of the Prior Arts

A wooden gate is made by assembling wooden boards, but the wooden gate may sag because the wooden boards will loosen after being used for a long time. As being sag, the wooden gate may deform and a gap may form between the wooden gate and a door frame, which makes the wooden gate unable to be closed tightly.

Therefore, a telescoping gate brace may be mounted on the wooden gate to prevent the wooden gate from sagging. Conventionally, a telescoping gate brace comprises two corner braces, a steel cable, and two clamps. When setting the telescoping gate brace, the two corner braces are mounted on two opposite corners of the wooden gate. Precisely, one of the corner braces is mounted on an upper corner that is close to the door frame, and the other corner brace is mounted on a lower corner that is away from the door frame. The steel cable is connected to the two corner braces, and at least one end of the steel cable is connected to the corner brace via a turnbuckle. To correspond to door gates of different dimensions, the steel cable can be folded in a suitable length, and then the two clamps secure the folded steel cable and thereby maintain the length. Finally, tighten the steel cable via adjusting the turnbuckle.

However, the conventional telescoping gate brace has the following defects:

1. The installation process takes too much time.
2. The strength is weak, so the useful lifetime is short.
3. The components of the telescoping gate brace are hard to secure with each other.
4. The maximum loading is not enough and thus the conventional telescoping gate brace cannot be adapted to a heavy wooden gate.

To overcome the shortcomings, the present invention provides a telescoping gate brace to mitigate or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

The main objective of the present invention is to provide a telescoping gate brace that is configured to be mounted on a wooden gate. The telescoping gate brace can bear a heavier wooden gate and has a more simple installation process.

The telescoping gate brace has a first rod, a second rod, a securing component, a first fixing assembly, a connector, and a second fixing assembly. The first rod has a first end and the second rod has a second end. The first rod and the second rod are movably sleeved with each other. The first end is away from the second rod and the second end is away from the first rod. The securing component is mounted on the first rod and the second rod, thereby securing a total length of the first rod and the second rod. The first fixing assembly is mounted on the first end and configured to be fixed on one of corners of the wooden gate. The connector is mounted on the second end. The second fixing assembly is mounted on the connector and configured to be mounted on another corner of the

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wooden gate. A distance between the second end and the second fixing assembly is adjustable via the connector.

With the first rod and the second rod sleeved with each other, the total length of the telescoping gate brace is easy to be adjusted, which also decreases the difficulty of the installation process and thus the installation is easier. Besides, the strength of the telescoping gate brace is improved, and thus the useful lifetime of the telescoping gate brace is longer and the telescoping gate brace can be adapted to a heavier wooden gate.

Other objectives, advantages and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a telescoping gate brace in accordance with the present invention;

FIG. 2 is an operational schematic view of the telescoping gate brace in FIG. 1, shown with a wooden gate;

FIG. 3 is an enlarged view of a first end of the telescoping gate brace in FIG. 1;

FIG. 4 is an exploded perspective view of the first end of the telescoping gate brace in FIG. 3;

FIG. 5 is an enlarged view of a second end of the telescoping gate brace in FIG. 1;

FIG. 6 is an exploded perspective view of the second end of the telescoping gate brace in FIG. 5; and

FIG. 7 is an enlarged view of a third mounting component of the telescoping gate brace in FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Please refer to FIG. 1 and FIG. 2. A telescoping gate brace is provided in accordance with the present invention and configured to be mounted on a wooden gate and thereby support the wooden gate. The wooden gate has two corners opposite each other, one is an upper corner and the other one is a lower corner.

Please refer to FIG. 3 and FIG. 4. The telescoping gate brace comprises a first rod 11, a second rod 12, a sleeve 13, a securing component 14, a first fixing assembly 20, a connector 30, and a second fixing assembly 40.

The first rod 11 has a first end and the second rod 12 has a second end. The first rod 11 and the second rod 12 are movably sleeved with each other, and the first end is away from the second rod 12 and the second end is away from the first rod 11. In this embodiment, the first rod 11 is sleeved in the second rod 12. It is not limited thereto, and the second rod 12 may be sleeved in the first rod 11 in another embodiment.

In this embodiment, the second rod 12 is hollow and thus the first rod 11 is movably mounted in the second rod 12. The first end of the first rod 11 is exposed from the second rod 12. Therefore, a total length of the first rod 11 and the second rod 12 is defined from the first end to the second end. In this embodiment, the first rod 11 has multiple first holes 110 and the second rod 12 has a second hole 120. The first holes 110 are arranged in a line parallel with the total length. After the first rod 11 is moved with respect to the second rod 12, one of the first holes 110 can align to the second hole 120.

The sleeve 13 is mounted on an end of the second rod 12. Said end is opposite the second end. The sleeve 13 surrounds the first rod 11. Therefore, at said end, the sleeve 13 may be

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located between the first rod **11** and the second rod **12** such that the first rod **11** may not contact the second rod **12**. The sleeve **13** has a third hole **130** and the third hole **130** is aligned to the second hole **120** of the second rod **12**. The first rod **11** and the second rod **12** may be made from metal, and the sleeve **13** may be made from plastic.

The securing component **14** is mounted on the first rod **11** and the second rod **12**, thereby securing the total length of the first rod **11** and the second rod **12**. Precisely, the securing component **14** is mounted in said one of the first holes **110**, the second hole **120**, and the third hole **130** and thereby the first rod **11** cannot move with respect to the second rod **12**, which secures the total length of the first rod **11** and the second rod **12**.

The first fixing assembly **20** is mounted on the first end and configured to be fixed on one of the corners of the wooden gate. In this embodiment, the first fixing assembly **20** comprises a first mounting component **21**, a positioning component **22**, and multiple first fixing components **23**. The first mounting component **21** has multiple mounting holes **210**. The mounting holes **210** are arranged in a line parallel with the first rod **11**. After the first mounting component **21** is moved with respect to the first rod **11**, one of the mounting holes **210** can align to the first hole **110**. The positioning component **22** is mounted in one of the mounting holes **210** and one of the first holes **110** and thereby the first mounting component **21** cannot move with respect to the first rod **11**, which secures the first rod **11** and the first mounting component **21**.

A distance between adjacent two of the first holes **110** differs from a distance between adjacent two of the mounting holes **210**. Therefore, to adjust the total length of the telescoping gate brace, moving the first rod **11** with respect to the second rod **12** and moving the first fixing assembly **20** with respect to the first rod **11** provide different effects.

The first fixing components **23** are mounted on the first mounting component **21** and configured to be mounted on the lower corner of the wooden gate. The first fixing components **23** are arranged in a line that is parallel with the first rod **11**, in other words, parallel with a length direction of the telescoping gate brace. In another embodiment, it is not limited thereto, and the first fixing components **23** are arranged in a line that is perpendicular to the first rod **11**.

Please refer to FIG. **5** and FIG. **6**. The second fixing assembly **40** comprises a second mounting component **41** and multiple second fixing components **42**. The second mounting component **41** is connected to the connector **30**. The second fixing components **42** are mounted on the second mounting component **41** and configured to be mounted on the upper corner of the wooden gate. The second fixing components **42** are arranged in a line that is perpendicular to the second rod **12**, in other words, perpendicular to the length direction of the telescoping gate brace. In another embodiment, it is not limited thereto, and the second fixing components **42** are arranged in a line that is parallel with the second rod **12**.

The connector **30** is mounted on the second end of the second rod **12** and the second fixing assembly **40** is mounted on the connector **30**. The second fixing assembly **40** is configured to be fixed on the other corner of the wooden gate. A distance between the second end and the second fixing assembly **40** is adjustable via the connector **30** and corresponds to the distance between the two corners of the wooden gate.

Precisely, the connector **30** comprises a fixed screwed rod **31**, a detachable screwed rod **32**, and a roller **33**. The fixed screwed rod **31** is mounted on the second end. The detach-

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able screwed rod **32** is mounted on the second fixing assembly **40** detachably. In this embodiment, the fixed screwed rod **31** is a straight rod and the detachable screwed rod **32** has a hook portion **321** and a straight portion **322**, and a thread is formed on the straight portion **322**.

The roller **33** is screwed on the fixed screwed rod **31** and the straight portion **322** of the detachable screwed rod **32**. A distance between the fixed screwed rod **31** and the detachable screwed rod **32** is adjustable via rolling the roller **33**, which also adjusts the distance between the second end and the second fixing assembly **40**. In this embodiment, the roller **33** may be polygonal such that the roller **33** can be rolled by a wrench.

Then please refer to FIG. **1** and FIG. **7**. The telescoping gate brace further has a third mounting component **15**. The third mounting component **15** is mounted on the second rod **12**, but it is not limited thereto; in another embodiment, the third mounting component **15** may be mounted on the first rod **11**. The third mounting component **15** is configured to be fixed on the wooden gate and enhances the resistance to the lateral forces.

Please refer to FIG. **1** and FIG. **2** again. The telescoping gate brace can be installed through the following steps. Originally, the first fixing assembly **20** is already mounted on the first rod **11** and the connector **30** is already mounted on the second rod **12**.

First, the first rod **11** or the second rod **12** is moved with respect to each other according to the diagonal length of the wooden gate and then the total length of the first rod **11** and the second rod **12** is secured by the securing component **14**. If a total length of the telescoping gate brace cannot correspond to the diagonal length of the wooden gate, the first mounting component **21** can be moved with respect to the first rod **11** to adjust the total length of the telescoping gate brace.

Second, the second fixing assembly **40** is mounted on the upper corner of the wooden gate. Then, the connector **30** is mounted on the second fixing assembly **40** with the first rod **11** and the second rod **12**.

Third, the first fixing assembly **20** is mounted on the lower corner of the wooden gate with the first rod **11** and the second rod **12**. Thus, the total length of the telescoping gate brace is secured.

In another embodiment, the first fixing assembly **20** may be mounted on the upper corner and the second fixing assembly **40** may be mounted on the lower corner of the wooden gate. The first fixing components **23** and the second fixing components **42** may be self-tapping screws, but it is not limited thereto.

Third, the connector **30** is adjusted to tighten the telescoping gate brace. In this embodiment, the roller **33** is rolled by a wrench to tighten the telescoping gate brace and lift and support the wooden gate.

Consequently, with the aforementioned structures and processes, the present invention provides the following advantages:

The first rod **11** and the second rod **12** are sleeved with each other such that the total length of the telescoping gate brace is easy to be adjusted.

The adjustment is simplified, which also decreases the difficulty of the installation process and thus the installation is easier.

With the sleeve **13** sleeved on the second rod **12** and surrounding the first rod **11**, noise generated during friction between the first rod **11** and the second rod **12** is decreased.

With the first fixing components **23** arranged in a line parallel with the first rod **11** or the second fixing component

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42 arranged in a line perpendicular to the second rod 12, the strength of the fixing portions that are connected to the wooden gate is improved.

The strength of the telescoping gate brace is improved, and thus the useful lifetime of the telescoping gate brace is longer and the telescoping gate brace can be adapted to a heavier wooden gate.

Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

What is claimed is:

1. A telescoping gate brace configured to be mounted on a wooden gate, the wooden gate having two corners opposite each other; the telescoping gate brace comprising:
 - a first rod having a first end;
 - a second rod having a second end, the first rod and the second rod movably sleeved with each other, and the first end being away from the second rod and the second end being away from the first rod;
 - a securing component mounted on the first rod and the second rod, thereby securing a total length of the first rod and the second rod;
 - a first fixing assembly mounted on the first end, wherein the first fixing assembly includes a separate mounting component configured to be fixed on one of the corners of the wooden gate by a separate fixing component;
 - a connector mounted on the second end; and
 - a second fixing assembly mounted on the connector, and configured to be mounted on the other corner of the wooden gate; a distance between the second end and the second fixing assembly being adjustable via the connector.
2. The telescoping gate brace as claimed in claim 1, wherein the connector comprises:
 - a fixed screwed rod mounted on the second end;
 - a detachable screwed rod detachably mounted on the second fixing assembly; and
 - a roller screwed on the fixed screwed rod and the detachable screwed rod; wherein a distance between the fixed screwed rod and the detachable screwed rod is adjustable via rolling the roller.
3. The telescoping gate brace as claimed in claim 2, wherein:
 - the second rod is a hollow rod and the first rod is sleeved in the second rod;
 - the first rod has:
 - a plurality of first holes arranged in a line;
 - the second rod has:
 - a second hole aligned to one of the first holes;
 - the securing component is mounted in said one of the first holes and the second hole thereby securing the total length of the first rod and the second rod.
4. The telescoping gate brace as claimed in claim 3 further comprising:
 - a sleeve mounted on an end of the second rod, said end opposite the second end, and the sleeve surrounding the first rod.
5. The telescoping gate brace as claimed in claim 4, wherein the sleeve forms:
 - a third hole aligned to the second hole.

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6. The telescoping gate brace as claimed in claim 5 further comprising:

a mounting component mounted on the second rod.

7. The telescoping gate brace as claimed in claim 4, wherein the first fixing assembly comprises:

a mounting component having a plurality of mounting holes arranged in a line; and

a positioning component mounted in one of the mounting holes and mounted in one of the first holes, the positioning component securing the total length of the first rod and the mounting component.

8. The telescoping gate brace as claimed in claim 7, wherein a distance between two of the adjacent first holes differs from a distance between two of the adjacent mounting holes.

9. The telescoping gate brace as claimed in claim 8, wherein the first fixing assembly comprises:

a plurality of fixing components mounted on the mounting component and configured to be mounted on the wooden gate; the fixing components of the first fixing assembly arranged in a line that is parallel with the first rod.

10. The telescoping gate brace as claimed in claim 9, wherein the second fixing assembly comprises:

a mounting component connected to the connector; and a plurality of fixing components mounted on the mounting component and configured to be mounted on the wooden gate; the fixing components of the second fixing assembly arranged in a line that is perpendicular to the second rod.

11. The telescoping gate brace as claimed in claim 1, wherein:

the second rod is a hollow rod and the first rod is sleeved in the second rod;

the first rod has:

a plurality of first holes arranged in a line;

the second rod has:

a second hole aligned to one of the first holes;

the securing component is mounted in said one of the first holes and the second hole, thereby securing the total length of the first rod and the second rod.

12. The telescoping gate brace as claimed in claim 11 further comprising:

a sleeve mounted on an end of the second rod, said end opposite the second end, and the sleeve surrounding the first rod.

13. The telescoping gate brace as claimed in claim 12, wherein the sleeve forms:

a third hole aligned to the second hole.

14. The telescoping gate brace as claimed in claim 1 further comprising:

a mounting component mounted on the second rod.

15. The telescoping gate brace as claimed in claim 1, wherein:

the first rod has a plurality of first holes arranged in a line; the first fixing assembly comprises:

a mounting component having a plurality of mounting holes arranged in a line; and

a positioning component mounted in one of the mounting holes and mounted in one of the first holes, the positioning component securing the total length of the first rod and the second rod.

16. The telescoping gate brace as claimed in claim 15, wherein a distance between two of the adjacent first holes differs from a distance between two of the adjacent mounting holes.

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17. The telescoping gate brace as claimed in claim 15, wherein the first fixing assembly comprises:

a plurality of fixing components mounted on the mounting component and configured to be mounted on the wooden gate; the fixing components of the first fixing assembly arranged in a line that is parallel with the first rod.

18. The telescoping gate brace as claimed in claim 1, wherein the second fixing assembly comprises:

a mounting component connected to the connector; and a plurality of fixing components mounted on the mounting component and configured to be mounted on the wooden gate; the fixing components of the second fixing assembly arranged in a line that is perpendicular to the second rod.

19. A telescoping gate brace configured to be mounted on a wooden gate, the wooden gate having two corners opposite each other, the telescoping gate brace comprising:

a first rod having a first end including a plurality of first holes;

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a second rod having a second end, the first rod and the second rod movably sleeved with each other, and the first end being away from the second rod and the second end being away from the first rod;

a securing component mounted on the first rod and the second rod, thereby securing a total length of the first rod and the second rod;

a first fixing assembly mounted on the first end, wherein the first fixing assembly includes a separate mounting component configured to be fixed on one of the corners of the wooden gate, the mounting component includes a plurality of second holes, and a distance between two adjacent first holes differs from a distance between two adjacent second holes;

a connector mounted on the second end; and

a second fixing assembly mounted on the connector and configured to be mounted on the other corner of the wooden gate, a distance between the second end and the second fixing assembly being adjustable via the connector.

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